

FINITENESS OF PLANAR FIVE-BODY CENTRAL CONFIGURATIONS FOR GENERIC EQUAL-PAIR MASSES

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ABSTRACT. Albouy and Kaloshin proved that the planar five-body problem has finitely many central configurations for all masses outside a codimension-two exceptional set. One component of that set is the equal-pair family $m_1 = m_2, m_3 = m_4$. We prove that for generic masses (a, a, b, b, c) in this family, that is, for all positive triples (a, b, c) outside a proper algebraic hypersurface, there are again only finitely many central configurations up to similarity. The proof is tropical. For each fixed value of the potential, the tropical variety of the mutual-distance equations lies in a polyhedral fan with 16,438 nonzero cones. Exact certificates (Laurent unit identities, explicit eliminations, and a modular criterion applied to whole stars of cones) exclude all but 98 of them, and these 98 lie in an open half-space. This forces finiteness for each potential value, and only finitely many potential values occur. All certificates can be checked with a standalone exact-arithmetic verifier.

1. INTRODUCTION

A configuration of n point masses $m_i > 0$ at positions $x_i \in \mathbb{R}^2$ is a *central configuration* (CC) if

$$\sum_{j \neq i} \frac{m_j (x_j - x_i)}{|x_j - x_i|^3} = -\lambda (x_i - x_{\text{cm}}) \quad \text{for all } i,$$

where x_{cm} is the center of mass and $\lambda > 0$. Smale's sixth problem asks whether, for every choice of positive masses, there are only finitely many CCs up to similarity. Hampton and Moeckel settled the case $n = 4$ [2]. Albouy and Kaloshin [1] settled $n = 5$ for all masses outside an explicit codimension-two subvariety $\mathcal{S}$ of the mass space. The equal-pair locus $m_1 = m_2, m_3 = m_4$ is one of its components: see Section 8.6, equation (32), and the proof of Theorem 6(ii) in [1].

Jensen and Leykin [3] gave a tropical proof for generic masses in the ambient mass space; their experiments with equal-pair mass valuations (Section 4.2 of [3]) did not lead to a finiteness proof. Equal valuations only fix the leading order of the masses. Here we instead impose the identities $m_1 = m_2, m_3 = m_4$ exactly, over a trivially valued coefficient field, and add the virial and potential equations at a fixed potential value.

Several earlier results treat this family under additional hypotheses. Moczurad and Zgliczyński [5] certify counts at selected exceptional mass choices, some in the equal-pair family, and discuss persistence nearby. The equal-mass case was classified in [7], following [8]. Kite [10], symmetric [11], trapezoidal [12], and convex [13, 14] configurations have been studied under geometric restrictions. Hampton and Jensen [9] proved spatial five-body finiteness with explicit mass exceptions, and Chang and Chen [15, 16, 17] developed symbolic diagram methods for six bodies; in [17, Section 10.6] a diagram survives exactly under equal-pair conditions equivalent to ours. Related work on certificate replication and on balanced configurations appears in [18] and [6].

The theorem below covers all real planar configurations, with no symmetry or shape assumption, for generic masses in the family. We do not determine the exceptional mass polynomial explicitly, and we do not count configurations.

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Outline of the proof.

For each fixed nonzero potential u , we study the augmented distance variety X'_u defined in Section 2. Its tropical variety is contained in a computable fan Σ with 16,438 nonzero cones. We prove finiteness in three steps.

- (1) *Exclude cones.* Polynomial identities, eliminations, and a modular criterion exclude 16,340 cones by showing that their initial equations have no torus zero (Sections 3–5).
- (2) *Separate the rest.* Every nonzero vector in the remaining 98 cones lies in one open half-space. Lemma 8 then makes X'_u finite.
- (3) *Return to configurations.* Only finitely many potential values occur for real normalized central configurations (Lemma 9), so their distance tuples lie in a finite union of the varieties X'_u .

Lemma 5 shows that one proper mass hypersurface suffices for all u .

Theorem 1. *There is a nonempty Zariski-open subset $\mathcal{O} \subset \{(a, b, c) \in \mathbb{R}_{>0}^3\}$ such that for every $(a, b, c) \in \mathcal{O}$, the planar five-body problem with masses (a, a, b, b, c) has finitely many central configurations up to similarity. Here similarity identifies translations, rotations, reflections, and positive changes of scale; body labels are retained.*

2. EQUATIONS AND TROPICAL SET-UP

We use the mutual-distance coordinates r_{ij} , $1 \leq i < j \leq 5$, in lexicographic pair order, with $r_{ji} = r_{ij}$ and $r_{ii} = 0$. Normalize the multiplier to $\lambda = M = \sum_i m_i$ and set $S_{ik} = r_{ik}^{-3} - 1$ for $i \neq k$. The ordered distance equations are

$$(1) \quad A_{ij} = \sum_{k \neq i} m_k S_{ik} (r_{jk}^2 - r_{ik}^2 - r_{ij}^2) = 0, \quad i \neq j.$$

There are $5 \cdot 4 = 20$ equations: the first index specifies the force vector being tested. We also use the ten redundant symmetric equations $A_{ij} + A_{ji} = 0$, $i < j$, together with the five determinants

$$(2) \quad \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & r_{ij}^2 & r_{ik}^2 & r_{i\ell}^2 \\ 1 & r_{ij}^2 & 0 & r_{jk}^2 & r_{j\ell}^2 \\ 1 & r_{ik}^2 & r_{jk}^2 & 0 & r_{k\ell}^2 \\ 1 & r_{i\ell}^2 & r_{j\ell}^2 & r_{k\ell}^2 & 0 \end{pmatrix} = 0, \quad i < j < k < \ell.$$

We clear the denominators by multiplying A_{ij} by $\prod_{k \neq i} r_{ik}^3$ and each symmetric equation by the product of its two denominators. After removing common distance monomials and nonzero rational scalars, we obtain the stored system $F = (f_1, \dots, f_{35})$: first the twenty ordered equations, then the ten symmetric equations, and finally the determinants. The indices are lexicographic within each block. These are units on the distance torus; no mass factor is removed. The equations agree with [3, Section 2.2]. The symmetric sums are algebraically redundant, but their tropical hypersurfaces cut down the prevariety.

Lemma 2 (Physical meaning of the distance equations). *For positive masses and a collision-free real planar configuration, its positive mutual distances satisfy the twenty equations (1) if and only if the configuration is central with multiplier M . This includes collinear configurations. Every such distance tuple satisfies (2) and hence F .*

Proof. Put $q_i = \sum_{k \neq i} m_k S_{ik} (x_k - x_i)$. The cosine identity gives $A_{ij} = -2q_i \cdot (x_j - x_i)$. Since q_i lies in the real span of the vectors $x_j - x_i$, the vanishing of all A_{ij} forces $q_i = 0$; this holds also for

collinear configurations. Moreover,

$$q_i = \sum_{k \neq i} \frac{m_k(x_k - x_i)}{r_{ik}^3} + M(x_i - x_{\text{cm}}).$$

Thus $q_i = 0$ is exactly the normalized central-configuration equation, and the converse follows from the same identity. The determinants vanish because every four points in the plane are affinely dependent. $\square$

Each ordered or symmetric polynomial is homogeneous of degree one in the masses, while the determinants do not involve masses. We may therefore divide all masses by $c > 0$ and take $c = 1$. Write $k_0 = \mathbb{Q}(a, b)$, $K = \overline{k_0}$, with trivial valuation, and $X = V(F) \subset (K^*)^{10}$.

Fixed potential.

We apply the half-space argument at a fixed nonzero potential. Let $I = \sum_i m_i |x_i - x_{\text{cm}}|^2$ and $U = \sum_{i < j} m_i m_j / r_{ij}$. Then $MI = \sum_{i < j} m_i m_j r_{ij}^2$, and central configurations satisfy the virial identity $U = MI$. Define

$$G = \sum_{i < j} m_i m_j r_{ij}^2 - U,$$

$$H = \sum_{i < j} m_i m_j \prod_{(k,l) \neq (i,j)} r_{kl} - U \prod_{k < l} r_{kl}.$$

Here U is an additional torus coordinate. For $u \neq 0$, write G_u, H_u for $U = u$ and put $X'_u = V(F, G_u, H_u)$. Every real normalized central configuration of potential u lies in X'_u , and by Lemma 9 only finitely many values u occur. We prove finiteness of each X'_u , not of $V(F)$.

Prevariety fan.

Let Σ be the polyhedral fan computed by `gfan _tropicalintersection` [20] from the supports of $f_1, \dots, f_{35}$ (max convention). It has f-vector $(1, 1603, 6110, 6525, 2200)$, 16,438 nonzero cones and trivial lineality space. Thus $\text{Trop}(X) \subseteq \bigcap_i \text{Trop}(V(f_i)) = |\Sigma|$.

3. GEOMETRIC AND SPECIALIZATION LEMMAS

Throughout, $\text{in}_w f$ is the max-initial form: the sum of the terms of f whose exponents maximize $\langle w, \cdot \rangle$. $\text{Trop}(X)$ is the set of $w \in \mathbb{R}^{10}$ such that $\text{in}_w I(X)$ contains no monomial. By the Fundamental Theorem [4, Thm. 3.2.3] and the Structure Theorem [4, Thm. 3.3.5], this is the tropicalization over the trivially valued algebraically closed field K . We use the max convention throughout; the min convention in [4] transfers by $w \mapsto -w$. For a cone $C = \text{cone}(v_1, \dots, v_s) \in \Sigma$ we put $w_C = \sum_j v_j$.

Lemma 3 (Cell constancy). *Let f be a polynomial whose initial form is used, with cone rays extended by a zero U -coordinate when necessary, and let*

$$\mathcal{F}_f = \bigcap_j \arg \max_{m \in \text{supp } f} \langle v_j, m \rangle.$$

If $\mathcal{F}_f \neq \emptyset$, then $\text{in}_w f = \text{in}_{w_C} f$ for every w in the relative interior of C .

Proof. Write $w = \sum \lambda_j v_j$ with all $\lambda_j > 0$. For $m_0 \in \mathcal{F}_f$ and $m \in \text{supp } f$, each $\langle v_j, m_0 - m \rangle \geq 0$, so $\langle w, m_0 - m \rangle \geq 0$. Equality holds exactly when every summand vanishes, that is, exactly when $m \in \mathcal{F}_f$. $\square$

The verifier checks this hypothesis for every generator used in a certificate.

Lemma 4 (Exclusion by a unit certificate). *Let Φ be a polynomial system in n torus coordinates r , over $\mathbb{Q}[a, b]$, and let $C \subset \mathbb{R}^n$ be a rational polyhedral cone. Choose $\mathcal{P} \subseteq \Phi$ satisfying Lemma 3 on C . Let $\mathcal{H}$ be the common homogeneity space of the forms $\text{in}_{w_C} f$, $f \in \mathcal{P}$, and let $\mathcal{D} \subseteq \{1, \dots, n\}$*

be a coordinate index set such that $\mathcal{H} \rightarrow \mathbb{Q}^{\mathcal{D}}$ is surjective. Denote by $\bar{f}$ the form obtained by fixing $r_j = 1$ for $j \in \mathcal{D}$. Suppose

$$\sum_{f \in \mathcal{P}} h_f \bar{f} + h_0 \left(1 - \zeta \prod_{j \notin \mathcal{D}} r_j\right) = \ell, \quad 0 \neq \ell \in \mathbb{Q}[a, b],$$

where all cofactors are in $\mathbb{Q}[a, b][\langle r_j \rangle_{j \notin \mathcal{D}}, \zeta]$. Then $\text{relint}(C) \cap \text{Trop}(V(\Phi)) = \emptyset$ over K . The identity remains an exclusion after specializing masses whenever $\ell \neq 0$ and the used initial supports are preserved.

Proof. If the initial ideal at $w \in \text{relint}(C)$ contains no monomial, its subideal generated by the chosen initial forms is proper in $K[r^{\pm 1}]$. The Nullstellensatz gives a common torus zero z . Integer vectors in $\mathcal{H}$ act on these zeros by coordinatewise torus scaling. The surjective projection onto $\mathcal{D}$ supplies an integer matrix with nonzero determinant on those coordinates. Over the algebraically closed field K , the corresponding isogeny of tori is surjective, so a zero can be moved to the slice $r_j = 1$, $j \in \mathcal{D}$. Substituting this zero and $\zeta = (\prod_{j \notin \mathcal{D}} r_j)^{-1}$ into the identity gives $0 = \ell$, a contradiction. The same identity proves the specialized assertion. $\square$

We apply this lemma both in the ten distance coordinates and in the eleven coordinates (r, U) , extending each distance weight by a zero U -coordinate.

Lemma 5 (Spreading exclusions to mass parameters). *Consider finitely many initial systems over $\mathbb{Q}[a, b]$, each with no torus zero over $\mathbb{Q}(a, b)$. There is a nonzero polynomial $q \in \mathbb{Q}[a, b]$ such that all these systems have no torus zero after any characteristic-zero specialization with $q(a, b) \neq 0$. A single such q can also preserve every support of a specified finite polynomial system.*

Proof. For each initial system, add $1 - \zeta \prod r_j$ and apply the Nullstellensatz. The resulting ideal is the unit ideal over $\mathbb{Q}(a, b)$, and hence over $\mathbb{Q}(a, b)$ by faithful scalar extension. Clearing the finitely many mass denominators in a unit identity gives a nonzero polynomial $q_i \in \mathbb{Q}[a, b]$. The product of these q_i and all nonzero coefficient polynomials of the specified system is a suitable q . When $q \neq 0$, the identities specialize and no coefficient vanishes, so the supports are unchanged. $\square$

The explicit certificates for augmented systems treat U as a torus variable, and the structural and modular exclusions use U -free forms. In either case the lemma produces a mass polynomial that does not depend on u , so one open set of masses works for every $u \neq 0$. For positive masses and $u \neq 0$, the coefficients of G_u, H_u are nonzero, so their supports are preserved as well.

A single specialization can lose solutions that pass to a neighboring cone, so the modular criterion checks the whole star of the target cone.

Lemma 6 (Exclusion from one modular specialization). *Let $\Phi = F \cup \{G, H\}$ as defined in Section 2, with U the extra coordinate of weight 0. For a cone $D \in \Sigma$ let Φ_D be the set of $f \in \Phi$ that satisfy Lemma 3 on D and whose form $\text{in}_{w_D} f$ does not involve U . Let $C \in \Sigma$ with $F \subseteq \Phi_C$. Suppose that there are a prime p and $(a_0, b_0) \in \mathbb{Z}^2$ such that*

- (1) *every coefficient (an element of $\mathbb{Z}[a, b]$) of every form $\text{in}_{w_C} f$, $f \in \Phi_C$, is nonzero modulo p at (a_0, b_0) , and*
- (2) *for every cone $D \in \Sigma$ having C as a face (including $D = C$), the forms $\text{in}_{w_D} f$, $f \in \Phi_C \cap \Phi_D$, specialized at (a_0, b_0) and reduced modulo p , have no common zero in the torus over $\overline{\mathbb{F}}_p$.*

Then the forms $\text{in}_{w_C} f$, $f \in \Phi_C$, have no common zero in the torus over $K = \overline{\mathbb{Q}(a, b)}$. Consequently, the cone is excluded for every augmented fixed-potential variety X'_u , $u \neq 0$, at generic masses.

Proof. Write $h \in \mathbb{Z}[a, b]$ as $h = \sum \gamma_{ij} (a - a_0)^i (b - b_0)^j$ with $\gamma_{ij} \in \mathbb{Z}$, and put $\nu(h) = \min\{v_p(\gamma_{ij}) + i\sqrt{2} + j\sqrt{3}\}$. Since $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$, the minimum is attained by exactly one term, and this term is multiplicative. So ν extends to a valuation of $\mathbb{Q}(a, b)$. Its residue field

is $\mathbb{F}_p$, and $\nu(h) = 0$ iff $h(a_0, b_0) \not\equiv 0 \pmod{p}$. Extend ν to K (residue field $\overline{\mathbb{F}}_p$, value group $\Gamma \subseteq \mathbb{R}$ divisible), and fix a splitting $\lambda \mapsto t^\lambda$ of ν on Γ [4, Lemma 2.1.15].

Suppose that z is a common torus zero over K , and write $z = t^{-w}y$ with $w \in \Gamma^{11}$ and $\nu(y_i) = 0$. Because the forms do not involve U , we may replace the U -entry of w by 0. For $g = \text{in}_{w_C} f = \sum c_m r^m$ with $f \in \Phi_C$, condition (1) gives $\nu(c_m) = 0$ for every m . Multiplying $g(z) = 0$ by $t^{h_{\max}}$, where $h_{\max} = \max_m \langle w, m \rangle$, and reducing gives $\overline{\text{in}_w g}(\bar{y}) = 0$. Here the bar denotes specialization at (a_0, b_0) modulo p . For sufficiently small $\varepsilon > 0$, uniformly for the finitely many f , we have $\text{in}_w \text{in}_{w_C} f = \text{in}_{w'} f$ with $w' = w_C + \varepsilon w$. If $w' \notin |\Sigma|$, then some $f \in F \subseteq \Phi_C$ has a monomial $\text{in}_{w'} f$. Its specialized coefficient is nonzero, so it cannot vanish at $\bar{y}$, a contradiction. Otherwise $w' \in \text{relint } D$ for some $D \in \Sigma$. Since ε is small and $w_C \in \text{relint } C$, C is a face of D . For $f \in \Phi_C \cap \Phi_D$, Lemma 3 gives $\text{in}_{w'} f = \text{in}_{w_D} f$. Thus $\bar{y}$ is a common torus zero of the specialized forms $\text{in}_{w_D} f$, $f \in \Phi_C \cap \Phi_D$, contradicting (2). The final statement follows from the unit ideal over k_0 and Lemma 5; its generators do not involve U , so the mass condition is uniform in u . $\square$

For the underlying valuation and initial-degeneration principles, see [19, Theorem 5.6] and [4]. Section 5 describes how condition (2) is verified.

Remark 7 (The star condition is necessary). The polynomials $f_1 = x + y$ and $f_2 = x + (1 + a)y + 1$ have no common zero at $a = 0$, although every coefficient survives. For $a \neq 0$ they have the torus solution $(x, y) = (1/a, -1/a)$. At the neighboring weight $(1, 1)$, both specialized max-initial forms become $x + y$, whose torus zeros are detected by the star check.

Lemma 8 (Residual half-space implies finiteness). *Let $W \subset (K^*)^n$ be an algebraic variety over a trivially valued algebraically closed field of characteristic zero. Let Σ be a finite polyhedral fan with $\text{Trop}(W) \subseteq |\Sigma|$. Let $\mathcal{E}$ be a collection of nonzero cones such that $\text{relint}(E) \cap \text{Trop}(W) = \emptyset$ for every $E \in \mathcal{E}$. Suppose each cone of $\Sigma \setminus \mathcal{E}$ is generated by its listed rays, and let $\mathcal{R}$ be the rays of these remaining cones. If $\eta \in \mathbb{Z}^n$ satisfies $\langle \eta, v \rangle > 0$ for every $v \in \mathcal{R}$, then W is finite.*

Proof. Every nonzero point of a remaining cone is a nonnegative combination of its rays, so $\langle \eta, w \rangle > 0$ for every nonzero $w \in \text{Trop}(W)$. Let Y be an irreducible component and consider the monomial map $r \mapsto r^\eta$. The tropicalization of the closure of its image is the linear image $\langle \eta, \text{Trop}(Y) \rangle$ [4, Corollary 3.2.13]. A closed irreducible subset of the one-dimensional torus is either a point or the whole torus, with tropicalization $\{0\}$ or $\mathbb{R}$, respectively. The latter cannot be contained in $[0, \infty)$, so $\langle \eta, w \rangle = 0$ on $\text{Trop}(Y)$ and hence $\text{Trop}(Y) = \{0\}$. The dimension theorem [4, Theorem 3.3.5] gives $\dim Y = 0$, and there are finitely many components. $\square$

Lemma 9 (Finitely many physical potential values). *Fix positive real masses. Among collision-free real planar central configurations normalized by $\lambda = M$, the potential takes only finitely many values, all positive. Every such configuration satisfies $U = MI = \sum_{i < j} m_i m_j r_{ij}^2$.*

Proof. Work over $\mathbb{C}$ on the center-of-mass-zero position space, restricted to configurations with all squared mutual distances nonzero. Adjoin nonzero r_{ij} satisfying $r_{ij}^2 = (x_i - x_j) \cdot (x_i - x_j)$. This distance cover is smooth and étale over the open position space because each derivative $2r_{ij}$ is invertible. The physical configurations lie on its positive real distance branches.

On this smooth cover put $f = U + \frac{M}{2}I$. Differentiating U in the position directions shows that the normalized force equations are exactly $df = 0$. The center-of-mass constraint loses no equation: the weighted sum of the vectors $\sum_{j \neq i} m_j (x_j - x_i)/r_{ij}^3 + Mx_i$ is zero. Therefore a critical point on the constrained space satisfies the full force equations, and the normalized central configurations belong to the algebraic critical locus of f .

On every irreducible component of that locus, the differential of the restricted function f vanishes on the dense smooth locus. In characteristic zero, f is therefore constant. Dotting the force equations with $m_i x_i$ and summing gives $U = MI$, so $f = \frac{3}{2}U$ there. The critical locus has finitely many irreducible components and hence only finitely many potential values on its physical points.

Finally, $MI = \sum_{i < j} m_i m_j r_{ij}^2$, and positivity gives $U > 0$. This is the physical version of [1, Lemma 2]. $\square$

Lemma 10 (The scaling ray and other negative directions). *Fix positive real masses and $u \in \mathbb{C}^*$, and put $X_u = V(F \cup \{G_u\}) \subseteq (\mathbb{C}^*)^{10}$. If $w \in \mathbb{R}^{10}$ has all coordinates negative, then $\text{in}_w G_u = -u$ is a unit, so $w \notin \text{Trop}(X_u)$. In particular the scaling ray $\mathbb{R}_{>0}(-1, \dots, -1)$ (cell 79) is excluded for every X_u . More generally, a cone on whose relative interior $\text{in}_w G_u$ is a single fixed monomial is excluded (96 cells before overlap with other exclusions).*

Proof. If all coordinates of w are negative, the monomials of G_u have w -weights $2w_{ik} < 0$, while the constant has weight 0; hence the constant is the unique initial term and is a unit. The same observation applies on any cone where the initial form is one fixed monomial. $\square$

Remark 11 (Certificates for the augmented system). Put $Y = V(F \cup \{G\}) \subseteq (\mathbb{C}^*)^{11}$. For $u \neq 0$, the embedding $r \mapsto (r, u)$ gives $\text{Trop}(X_u) \times \{0\} \subseteq \text{Trop}(Y)$, since the valuation is trivial. A unit certificate excluding $(w, 0)$ from $\text{Trop}(Y)$ therefore excludes w from every $\text{Trop}(X_u)$. The checker applies Lemmas 3 and 4 to cone rays extended by a zero U -coordinate and verifies initial-form constancy on $C \times \{0\}$.

Remark 12 (Certificates using the definition of U). The polynomial H in Section 2 is the denominator-cleared definition of the potential. Put $Y' = V(F \cup \{G, H\}) \subseteq (\mathbb{C}^*)^{11}$. The same embedding gives $\text{Trop}(X'_u) \times \{0\} \subseteq \text{Trop}(Y')$, while $X'_u \subseteq X_u$ gives $\text{Trop}(X'_u) \subseteq \text{Trop}(X_u)$. Thus certificates for F , $F \cup \{G\}$, or $F \cup \{G, H\}$ all apply to X'_u .

Proof of Theorem 1 from the certificates. We first work at generic masses over $\overline{\mathbb{Q}(a, b)}$. The identities, elimination replays, and Lemma 6 exclude the certified cones for every X'_u , $u \neq 0$.

Lemma 5, together with the denominators of the explicit identities and the coefficient polynomials of F , gives a hypersurface $\mathcal{Z} = V(q) \subset \mathbb{C}^2$, with $0 \neq q \in \mathbb{Q}[a, b]$. We enlarge $\mathcal{Z}$, if necessary, by multiplying q by $R_{325}(a, b)R_{325}(b, a)$, where R_{325} is defined in (3). Outside $\mathcal{Z}$, the support fan and all exclusions remain valid for every $u \neq 0$.

The coverage and half-space checks in Section 4 therefore make every X'_u finite by Lemma 8. Lemma 9 supplies finitely many positive potential values for the real central configurations, so their distance tuples lie in a finite union of these varieties. A labeled real distance tuple determines a configuration up to Euclidean isometry, including reflection, and $\lambda = M$ fixes scale. Finally, a nonzero polynomial cannot vanish on the entire positive quadrant, so $\mathbb{R}_{>0}^2 \setminus \mathcal{Z}$ is nonempty. Pulling it back under $(a, b, c) \mapsto (a/c, b/c)$ and clearing powers of c gives $\mathcal{O} = \{(a, b, c) \in \mathbb{R}_{>0}^3 : Q_*(a, b, c) \neq 0\}$ for a nonzero homogeneous $Q_* \in \mathbb{Q}[a, b, c]$. Thus $\mathcal{O}$ is the positive-mass complement of a hypersurface defined over $\mathbb{Q}$. $\square$

Remark 13. The point $(a, b) = (3, 7)$ in Lemma 6 is used only to certify generic exclusions. Some explicit identities have denominators vanishing at $(3, 7)$, so the argument does not prove finiteness at those particular masses.

4. COMPUTATION

Symmetry.

The swaps (1 2) and (3 4) preserve both the polynomial system and the fan, giving 4,781 orbits under this order-four group of the 16,438 nonzero cones. The relabeling $\sigma = (1 3)(2 4)$, together with $a \leftrightarrow b$, supplies an additional symmetry, and the generated group has order eight. The verifier checks these actions on the polynomials and on the fan, and rechecks every transported identity at its target cone.

Search and exact certificates.

A fast unit-ideal test at random masses modulo 32003 was used to choose subsets of initial forms; it plays no role in the proof. SINGULAR [21] then computed unit ideals over $\mathbb{Q}(a, b)$ and the corresponding lift cofactors. Some computations became tractable only after removing common monomials, allowing a power of the coordinate product on the right-hand side, or halving exponents of variables that occur only to even powers. The remaining cases were handled by split trees or explicit eliminations. Section 5 describes these formats.

The companion ledger.

An independently generated ledger of fourteen batches of Laurent identities records generator indices, cell weights, monomial shifts, multipliers, and a nonzero mass factor q . Each identity has the form

$$\sum_j c_j r^{-e_j} \text{in}_{w_C} f_{i_j} = q(a, b, c) r^d.$$

The checker expands both sides exactly, verifies that the recorded weight is the sum of the cell rays, and checks support constancy. The right-hand side is a Laurent unit at generic masses. The ledger also includes the structural proof for cell 346 and its symmetry images, conditional transfers from cell 325, and coefficient-level symmetry transports. Together these exclude 10,965 cells.

For cell 325, the ratio elimination gives the necessary mass condition

$$(3) \quad R_{325}(a, b) = (16a^2 - b^2)(4a^2 + b^2) = 0.$$

Thus $R_{325}(a, b) \neq 0$ excludes its initial system over the algebraic closure. Mass exchange gives the condition $R_{325}(b, a) \neq 0$; for positive masses the exceptional lines are $b = 4a$ and $a = 4b$. Both polynomials are included in $\mathcal{Z}$ in the proof of Theorem 1.

Lemma 14 (Complex exclusion for the companion structural cell). *The initial-generator system for cell 346 has no common torus zero over $\overline{\mathbb{Q}(a, b)}$ (with $c = 1$). It also has no common complex torus zero for any fixed positive real masses a, b, c .*

Proof. Eliminating the distance ratios $t = r_{25}/r_{15}$ and $h = r_{12}/r_{15}$ gives the necessary mass condition

$$(a - 4c)(a + 4c)(a^2 + 4c^2)^2(a^4 + 4a^3c + 6a^2c^2 + 3ac^3 + c^4)c^8P(a/c) = 0,$$

where $P(z) = z^8 - z^7 + z^6 - z^4 + 8z^3 + 3z^2 - 3z + 1$. This is a nonzero polynomial, so it excludes complex torus zeros at generic masses without any positivity assumption. For positive masses, a Sturm sequence shows that P has no positive root, so only $a = 4c$ remains. The algebraic distance equations then force $t = 1$, $h = -2$, and the necessary condition $(b^2 - 256c^2)(b^2 + 32c^2)^2 = 0$. Positivity leaves $b = 16c$. On this branch, polynomial reduction over $\mathbb{Q}$ forces $r_{14} = r_{13}$ and $r_{34} = -2r_{13}$, while another initial generator becomes $-72cr_{13}^4r_{35}^2$, a nonzero torus monomial. All distance coordinates in this argument are complex. The exclusion transfers to the symmetry images of cell 346. $\square$

The final three exclusions.

Representatives 81, 121, and 206 use Lemma 6 at $(a, b) = (3, 7)$, $p = 32003$. Each checked star has 60 cones, including its target. Mass exchange also excludes partners 316, 50, and 220.

Coverage and residual.

Table 1 gives disjoint counts by assigning each cone to the first applicable certificate in the checked ledger. Symmetry images under the pair swaps are included in their source rows, while the additional transport row records the remaining four mass-exchange images. The 96 single-term virial cells overlap earlier rows, leaving 57 in that row and the separate scaling ray.

TABLE 1. Disjoint accounting of the nonzero support-fan cones.

Certificate or residual class	Cones
Companion Laurent and structural identities	10,965
Other Laurent identities	5,258
Resultant specialization	8
Split trees	6
Structural elimination	25
Modular star criterion	16
Additional mass-exchange transport	4
Single-term virial initial forms	57
Scaling ray	1
Excluded total	16,340
Residual	98
Total	16,438

TABLE 2. Certificate representatives and expansion for Table 1.

Class	Representatives and check in Section 5	Cones
Companion	Fourteen batches, structural and conditional transfers; companion ledger above	10,965
Laurent	1,562 representatives; unit identities	5,258
Resultant	227, 2440; two-variable resultant specialization	8
Split trees	378, 427; binary split identities	6
Elimination	334, 1767; 49, 2171, 2174, 2175; mass-exchange partners 2345, 218, 2885, 3494, 3545	25
Modular stars	81, 121, 206; partners 316, 50, 220	16
Transport	Remaining mass-exchange images from 353, 3097	4
Single-term G	23 representatives; enumerated accounting and Lemma 10	57
Scaling ray	79; Lemma 10	1

Pair swaps give orbits of sizes one, two, or four. The split representatives contribute $2 + 4$ cones and the resultants $4 + 4$. In the elimination row, 334 contributes one cone; 49, 218, 1767, 2174, 2175, 2345, 3494, and 3545 contribute two each; and 2171 and 2885 contribute four each. The six modular representatives contribute $2 + 4 + 2 + 2 + 4 + 2$.

The residual set of 98 cones is invariant under the symmetry group. It has 77 distinct primitive rays, and every one has positive pairing with

$$(4) \quad \eta = (-23, 1, 1, 51, 1, 1, 51, 24, 13, 13).$$

The checker reconstructs the residual from the certificate coverage and verifies these pairings in integer arithmetic.

5. CERTIFICATES AND THEIR VERIFICATION

This section describes the certificate formats and what the verifier checks for each. File names and commands are listed in Section 6.

Laurent unit certificates.

A certificate for a cone C records the subset $\mathcal{P}$ of the system, the kept coordinates, and for each $g \in \mathcal{P}$ a cofactor C_g with coefficients in $\mathbb{Q}(a, b)$. The verifier, written with python-flint [22], parses

the fan and the polynomial system directly. It checks the hypothesis of Lemma 3 for each $g \in \mathcal{P}$, recomputes $\mathcal{H}$ and checks that it surjects onto the fixed coordinates $\mathcal{D}$, clears denominators into $\ell = \text{lcm}$, and verifies

$$\sum (\ell C_g) \bar{g} + (\ell C_0) (1 - \zeta \prod r) = \ell$$

in exact arithmetic. Denominators lie in $\mathbb{Z}[a, b]$, so the cleared identity specializes wherever $\ell(a, b) \neq 0$. Four variants occur.

- *Augmented systems.* The certificate uses $F \cup \{G\}$ or $F \cup \{G, H\}$ in the eleven coordinates (r, U) , with U of weight 0 (Remarks 11 and 12). This covers the scaling ray, the cones on which $\text{in}_w G$ is a single term (with the one-term certificate $C_G = 1/\text{coeff}$), and cones whose unit test without G did not finish.
- *Monomial division.* Each $\bar{g}$ is first divided by the greatest common monomial of its terms. This does not change its torus zero set.
- *Even exponents.* If a variable r_v occurs in every $\bar{g}$ only to even powers, write $\bar{g}(r) = \bar{g}'(r_v^2)$. Since $r_v \mapsto r_v^2$ is surjective on the torus, an identity $\sum C'_i \bar{g}'_i = s^N$ pulls back to $\sum C'_i(r^2) \bar{g}_i = r^{2N}$.
- *Monomial right-hand side.* The identity $\sum_g (\ell C_g) \bar{g} = \ell (\prod_{j \in \mathcal{I}} r_j)^N$, where $\mathcal{I}$ is the set of kept coordinates, excludes torus zeros without a Rabinowitsch variable.

Dehomogenized computations.

When every generator in $\mathcal{P}$ is homogeneous in (a, b) , we compute cofactors over $\mathbb{Q}(a)$ at $b = 1$. If $\sum_i C_i(a) \bar{g}_i(a, 1) = 1$ and $\bar{g}_i$ has degree d_i in (a, b) , then $\sum_i b^{-d_i} C_i(a/b) \bar{g}_i(a, b) = 1$. The resulting certificate is an ordinary identity over $\mathbb{Q}(a, b)$ and is checked as such.

Split trees.

For a few cones a single lift identity is too large to compute, but the ideal splits. Typically one initial form factors modulo a binomial of the system; for cone 378,

$$a^2 + ab r_{13}^2 + ab r_{24}^2 + b^2 r_{34}^2 \equiv (a + b r_{13}^2)(a + b r_{24}^2) \pmod{r_{34}^2 - r_{13}^2 r_{24}^2}.$$

A split-tree certificate is a finite binary tree. Each node v has the path ideal $\mathfrak{J}_v = \mathfrak{J} + (p_1, \dots, p_s)$, where $\mathfrak{J}$ is the Rabinowitsch ideal above and the p_i are the polynomials chosen at its ancestors. A leaf carries cofactors with $\sum_i C_i k_i = \ell_v \in \mathbb{Q}[a, b] \setminus \{0\}$. A split node carries q_1, q_2 and cofactors with $\sum_i C_i k_i = \ell_v q_1 q_2$, and both children $\mathfrak{J}_v + (q_1)$, $\mathfrak{J}_v + (q_2)$ must be certified. If $q_1 q_2 \in \mathfrak{J}_v$ and $1 = x_1 + c_1 q_1 = x_2 + c_2 q_2$ with $x_j \in \mathfrak{J}_v$, then $1 = (x_1 + c_1 q_1)(x_2 + c_2 q_2) \in \mathfrak{J}_v$. By induction $1 \in \mathfrak{J}$. The generic condition is that no ℓ_v vanishes.

Elimination certificates (cones 334, 1767 and 49, 2171, 2174, 2175).

In each group, six initial forms $g_1, \dots, g_6$ in five variables suffice, and the forms are the same at every cone of the group; they are listed in Appendix A. The argument is pointwise. Let z be a common torus zero of the g_i over K . Every derived polynomial below vanishes at z , and each branch ends in a contradiction.

- (1) Three of the forms are affine-linear in three squares. Cramer's rule expresses $D r^2 - N_r$ as a combination of them, with D a monomial. Substituting for the squares and squaring away the remaining linear unknowns gives division-free polynomials $E_1, \dots, E_{14}$ in two variables (x, y) .
- (2) We divide only by factors known to be nonzero at z : monomials, the masses, and the factors X_q, Y_q, S_q of the numerators N_r , which are nonzero because $D r^2 = N_r$ and $r \neq 0$. The factors F_i of the Cramer determinant of the second linear system are treated as separate branches $F_i(z) = 0$.
- (3) Resultants lie in the ideals of their arguments, so they vanish at z . The final non-vanishing $\text{Res}_y(R', R'') \neq 0$ is certified by one modular specialization at which the relevant degrees

are preserved and the gcd is 1. For 49 the remaining common factor is the irreducible cubic $\phi = (ay + 1)^3 - b^2(ay^3 + 1)$. Over $\mathbb{L} = \mathbb{Q}(a, b)[y]/(\phi)$ let $\psi = -\beta_0/\beta_1$, where $\beta_1x + \beta_0$ is the first subresultant of X_q, Y_q . The verifier checks that $X_q(\psi) = 0$, that $(x - \psi)^2$ divides $\widehat{E}_{10}$ and $\widehat{E}_{13}$ in $\mathbb{L}[x]$, and that $\deg \gcd_{\mathbb{L}}(\widehat{E}_{10}, \widehat{E}_{13}) \leq 2$ (Appendix A). Hence $x(z) = \psi$ and $X_q(z) = 0$, a contradiction.

For 334, all six forms are homogeneous of degree one in (a, b) , so we divide by b and replace a by a/b ; this is an equivalence, not a specialization. The diagonal $r_{23} = r_{24}$ is closed by a resultant chain in $\mathbb{Z}[a]$. Off the diagonal, the involution $(r_{13} r_{14})(r_{23} r_{24})$ preserves $g_1, \dots, g_5$, and g_6 enters only through norms $E \cdot \sigma E$, so we may work in $s = r_{23} + r_{24}$ and $\pi = r_{23}r_{24}$. The only remaining common zero, $\pi = 3/(a^2 - 1)$, $s = -3a/(a^2 - 1)$, has $X_q Y_q S_q = 0$, which is impossible on the torus.

Bivariate leaves.

Several branches end with finitely many bivariate polynomials. The verifier computes a nonzero resultant in x of two of them; every common root has y among its roots. For each irreducible factor it forms the corresponding extension field, specializes all the polynomials, and computes their gcd in x , removing only the explicitly excluded factors. A constant gcd in every fiber shows that no admissible common root exists.

Modular star certificates.

For the targets $k = 81, 121, 206$ the verifier computes, directly from the fan, the set of cones whose rays contain the rays of C . This contains every coface, and extra cones are harmless. It checks that all 37 initial forms are constant on C and free of U (stronger than the requirement $F \subseteq \Phi_C$), that every coefficient is nonzero modulo 32003 at $(3, 7)$, and condition (2) of Lemma 6 at every cone of the star, using the forms in $\Phi_C \cap \Phi_D$. Most neighboring cones have ordinary modular cofactor identities. The cones 2174, 2885, and 3494 instead repeat the elimination of cone group 49 modulo p , ending with fourteen bivariate polynomials whose common roots force a zero coordinate.

At the target cones themselves, three variables occur only to even powers and are replaced by their squares. One form eliminates a squared coordinate, and a second gives

$$c_{23}(x, y)s_{23} + c_{24}(x, y)s_{24} = 0.$$

Outside $c_{23} = c_{24} = 0$, put $s_{23} = \kappa P$, $s_{24} = \kappa Q$ with κ, P, Q and the eliminated coordinate nonzero. The remaining homogeneous equations no longer involve κ . Three of them determine S^2, T^2, W^2 , and two cubic relations make $1/T$ and $1/W$ affine in $1/S$; squaring the first relation then determines $1/S$. Every denominator is checked to be nonzero on its branch (for cell 206 this needs one more bivariate leaf), and four further leaves treat the cases in which a coefficient of a cubic relation vanishes. Substituting into further initial forms gives four bivariate polynomials, handled as above.

On the branch $c_{23} = c_{24} = 0$ one gets $(x^3, y^3) = (1, 1)$ for 81 and 121, and $(1, -1/13)$ for 206. Since $32003 \equiv 2 \pmod{3}$, each nonzero element of $\mathbb{F}_{32003}$ has a cube root there, and all nine root pairs lie in $\mathbb{F}_{32003}[t]/(t^2 + t + 1)$. At each of them a small cofactor identity shows that the remaining forms have no common zero.

The support fan.

Gfan [20] versions 0.7 and 0.6.2 produce the same rays and cones. The proof does not rely on gfan: two independent checks show that the stored cones are exactly the cones of the prevariety.

The first check concerns the structure of the fan. For each stored cone and each support, it intersects the maximizing faces at the rays of the cone, obtaining a face T_i with at least two terms. The corresponding normal region is given by equalities within T_i and the inequalities $\langle w, m_0 - m \rangle \geq 0$, $m_0 \in T_i$, for the other terms. Exact nullspace computations show that the solution space of the equalities is the span of the cone. Facet normals are computed from independent sets of $d - 1$ rays, and each facet inequality is shown to be among the defining inequalities, so the stored cone equals the normal region. The check also confirms that the rays are primitive and extreme,

that the lineality space is zero, and that every facet is a stored cone (44,998 incidences). Since cones of a common refinement of normal fans meet in common faces, the relative interiors partition the support. This is the face relation used in Lemma 6. The same check confirms that the 35 supports and the fan are invariant under the symmetry group.

The second check shows that the support is complete. Using cddlib with GMP [23], the 3,120 maximal cones are converted to exact half-space descriptions. A linear-arithmetic query then asks for a weight w outside their union that lies on all 35 tropical hypersurfaces: for each support $\mathcal{A}_i$ we introduce t_i with

$$t_i \geq \langle w, m \rangle \quad (m \in \mathcal{A}_i), \quad \#\{m \in \mathcal{A}_i : t_i = \langle w, m \rangle\} \geq 2.$$

By homogeneity we may assume $\max_j |w_j| = 1$. The order-eight symmetry group has three coordinate orbits, represented by r_{12}, r_{13}, r_{15} , so it suffices to fix one of these at ± 1 . The stabilizers allow further normalizations: the largest cross-pair coordinate can be placed at r_{13} when r_{12} is fixed, $w_{12} \geq w_{34}$ when r_{13} is fixed, and $w_{13} \geq w_{14}$ when r_{15} is fixed. All six cases are unsatisfiable in both Z3 [24] and Yices [25], using exact rational arithmetic and two independent encodings of the counting constraint. A nonempty difference of rational polyhedral sets contains a rational point, so this proves the containment for real weights. The solvers do not produce checkable proofs, so they are part of the trusted base.

Coverage.

The final check reconstructs the set of excluded cones from the certificates themselves, not from search output, takes its complement, and verifies the pairing with (4) on each of the 77 residual rays in integer arithmetic.

Trusted software.

The verification relies on Python, FLINT through python-flint, SymPy (for the companion ledger), and, for fan completeness, cddlib/GMP, Z3, and Yices. Gfan and SINGULAR produced the certificates but are not needed to check them. The proof has not been formalized in a proof assistant.

6. CODE AND DATA

The source code, verifier, and certificates are available at

<https://github.com/brandonlign/tropcc>

and archived at <https://doi.org/10.5281/zenodo.23197573>. The certificates for this paper form the archive attached to release v1.3.4, whose SHA-256 digests are listed in `certificate_manifest.json`. The README gives the commands for a full replay. The main components are as follows.

Component	Location
Polynomial system and fan	<code>data/sys.json, data/fan.out</code>
Equation reconstruction	<code>companion/checkers/verify_equations.py</code>
Laurent certificates	<code>data/exact/*cert/, verifier/verify.py</code>
Split trees	<code>data/exact/split/, verifier/verify_split.py</code>
Eliminations	<code>data/elimination_forms.json, verifier/verify_{334,49}.py</code>
Modular stars	<code>verifier/verify_lemS{,_rep}.py, b0_verify.py</code>
Companion ledger	<code>companion/results/</code>
Symmetry	<code>verifier/symcheck.py, symcheck2.py</code>
Fan structure, completeness	<code>verifier/fan_{structure,completeness,second_solver}.py</code>
Cone accounting	<code>data/cone_accounting.json</code>
Negative controls	<code>verifier/review_checks.py</code>

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APPENDIX A. EXPLICIT ELIMINATION INPUTS AND DERIVED POLYNOMIALS

This appendix gives the input forms and derived polynomials for the two elimination certificates of Section 5.

Cone group 334.

Put $(A, B, T, x, y) = (r_{13}, r_{14}, r_{34}, r_{23}, r_{24})$ and take $b = 1$ after the homogeneous mass reduction above. The homogeneous forms are recovered from the displayed normalized forms by $\tilde{g}_i(a, b; r) = b g_i(a/b; r)$. Thus the mass-independent coefficient terms acquire a factor b , while distance-slice constants retain their meaning. The six normalized forms, in their checked order, are

$$\begin{aligned}
g_1 &= -A^3 B^3 a - A^3 \\
&\quad + AB^3 a (x - 1) (x + 1) - AB^2 \\
&\quad + AT^2 - 2B^3 \\
g_2 &= -A^3 B^3 a + A^3 B a (y - 1) (y + 1) \\
&\quad - 2A^3 - A^2 B \\
&\quad - B^3 + BT^2 \\
g_3 &= A^2 y^3 + B^2 x^3 \\
&\quad - 2ax^3 y^3 - x^3 y^2 - x^3 - x^2 y^3 - y^3 \\
g_4 &= A^2 a x y^3 + T^2 x \\
&\quad - ax^3 y^3 - ax y^3 - x^3 - x y^2 - 2y^3 \\
g_5 &= B^2 a x^3 y + T^2 y \\
&\quad - ax^3 y^3 - ax^3 y - 2x^3 - x^2 y - y^3 \\
g_6 &= -2B^3 T^3 a - B^3 T^2 y \\
&\quad + B^3 y (x - y) (x + y) - B^2 T^3 a y \\
&\quad - T^3 a y (y - 1) (y + 1)
\end{aligned}$$

The same forms occur at cone 1767. The variables A, B, T in this appendix are local distance variables and are unrelated to A_{ij} .

Cone group 49.

Put $(A, B, T, x, y) = (r_{14}, r_{15}, r_{45}, r_{34}, r_{35})$. In this group both parameters remain. The six forms are

$$\begin{aligned}
g_1 &= -A^3 B^3 b - A^3 \\
&\quad + AB^3 b (x - 1) (x + 1) - AB^2 \\
&\quad + AT^2 - 2B^3 b \\
g_2 &= -A^3 B^3 b + A^3 B b (y - 1) (y + 1) \\
&\quad - 2A^3 - A^2 B b \\
&\quad - B^3 b + BT^2 b
\end{aligned}$$

$$\begin{aligned}
 g_3 &= A^2by^3 + B^2x^3 \\
 &\quad - 2ax^3y^3 - bx^2y^3 - by^3 - x^3y^2 - x^3 \\
 g_4 &= A^2axy^3 + T^2x \\
 &\quad - ax^3y^3 - axy^3 - 2by^3 - x^3 - xy^2 \\
 g_5 &= B^2ax^3y + T^2by \\
 &\quad - ax^3y^3 - ax^3y - bx^2y - by^3 - 2x^3 \\
 g_6 &= -2B^3T^3b - B^3T^2by \\
 &\quad + B^3by(x-y)(x+y) - B^2T^3ay \\
 &\quad - T^3ay(y-1)(y+1)
 \end{aligned}$$

These forms agree at cones 49, 2171, 2174, and 2175. The two groups use different initial subsets and distance slices, which explains the different mass coefficients in g_1, g_2, g_6 . Their forms g_3, g_4, g_5 agree at $b = 1$, so the X_q, Y_q, S_q used for group 334 are exactly the $b = 1$ specializations of those defined below for group 49.

A common division-free construction.

The two elimination replays share a division-free reduction. In either group write

$$(g_3, g_4, g_5)^\top = M(A^2, B^2, T^2)^\top - v,$$

where M, v are obtained by coefficient extraction from the displayed forms. Put $d = \det M$, and let n_A, n_B, n_T be its Cramer numerators. Thus

$$d(A^2, B^2, T^2)^\top - (n_A, n_B, n_T)^\top = \text{adj}(M)(g_3, g_4, g_5)^\top.$$

The determinant d is a nonzero monomial and hence a unit on the generic mass and distance torus. For a polynomial P , define $\rho(P)$ termwise by replacing each $A^{2i+e}B^{2j+f}T^{2k+h}$ with $n_A^i n_B^j n_T^k A^e B^f T^h d^{N-i-j-k}$, where $e, f, h \in \{0, 1\}$ and N is the largest $i + j + k$ among its terms. On a common zero, $\rho(P) = d^N P$. Let prim remove integer content and the greatest common monomial, including mass monomials. These removed factors are nonzero on the generic torus. Set $R_i = \text{prim } \rho(g_i)$ for $i = 1, 2, 6$, and extract coefficients by

$$\begin{aligned}
 R_1 &= A(a_1B + b_1) + c_1B, \\
 R_2 &= B(a_2A + c_2) + b_2A, \\
 R_6 &= T(a_6B + b_6) + c_6B.
 \end{aligned}$$

The lowercase coefficients here are polynomials in x, y over the mass ring, not mass parameters. Applying $\text{prim } \rho$ to $c_1^2B^2 - A^2(a_1B + b_1)^2$, $b_2^2A^2 - B^2(a_2A + c_2)^2$, and $c_6^2B^2 - T^2(a_6B + b_6)^2$ gives, respectively,

$$\alpha B + \beta, \quad \gamma A + \delta, \quad \alpha_6 B + \beta_6.$$

Next, take $\text{Res}_A(R_1, R_2)$ and $\text{Res}_B(R_1, R_2)$, remove their maximal factors B and A , respectively, and apply $\text{prim } \rho$. The results are $\mu B + \omega_0$ and $\mu_2 A + \omega_2$. Define the following polynomials, applying

prim to each right-hand side:

$$\begin{aligned}
E_1 &= \beta^2 d - \alpha^2 n_B, & E_2 &= \delta^2 d - \gamma^2 n_A, \\
E_3 &= \alpha_6 \beta - \alpha \beta_6, \\
E_4 &= a_1 \beta \delta - b_1 \alpha \delta - c_1 \gamma \beta, \\
E_5 &= a_2 \delta \beta - c_2 \beta \gamma - b_2 \delta \alpha, \\
E_6 &= (b_1 \delta)^2 d - (c_1 \gamma - a_1 \delta)^2 n_B, \\
E_7 &= (b_2 \delta)^2 d - (c_2 \gamma - a_2 \delta)^2 n_B, \\
E_8 &= (c_1 \beta)^2 d - (b_1 \alpha - a_1 \beta)^2 n_A, \\
E_9 &= (c_2 \beta)^2 d - (b_2 \alpha - a_2 \beta)^2 n_A, \\
E_{10} &= \omega_0^2 d - \mu^2 n_B, & E_{11} &= \mu \beta - \alpha \omega_0, \\
E_{12} &= \mu \beta_6 - \alpha_6 \omega_0, & E_{13} &= \omega_2^2 d - \mu_2^2 n_A, \\
E_{14} &= \mu_2 \delta - \gamma \omega_2.
\end{aligned}$$

Each E_i vanishes at a common torus zero. The constructions use multiplication, squaring, resultants, and division only by known units.

Factors and branches for group 49.

The following factors must be nonzero:

$$\begin{aligned}
X_q &= a^2 x^4 y^4 + abx^3 y^4 + abxy^4 + b^2 y^4 - x^4, \\
Y_q &= a^2 x^4 y^4 + ax^4 y^3 + ax^4 y - b^2 y^4 + x^4, \\
S_q &= a^2 x^4 y^4 - b^2 y^4 - bx^3 y - bxy^3 - x^4.
\end{aligned}$$

Indeed $n_A = -2x^3 X_q$, $n_B = -2by^3 Y_q$, $n_T = 2ax^3 y^3 S_q$. Remove their maximal powers from E_i to obtain $\tilde{E}_i$. On the leaf, also remove powers, while the quotient is nonconstant, of

$$ax^2 y^2 \pm by^2 \pm x^2 \quad (\text{all four sign choices}), \quad ax^3 y^3 + by^3 + x^3,$$

to obtain $\hat{E}_i$. The zero loci of these five factors are checked separately using $\tilde{E}_1, \tilde{E}_2, \tilde{E}_3$, so no division discards a branch. The leaf uses $\text{Res}_x(\hat{E}_{10}, \hat{E}_{13})$ and $\text{Res}_x(\hat{E}_{10}, \hat{E}_{12})$ as described in Section 5.

Irreducibility and the specialization bound.

Let $\phi = (ay+1)^3 - b^2(ay^3+1)$. Over $\mathbb{Q}(a, y)$ the quotient $(ay+1)^3/(ay^3+1)$ is not a square: its order at $y = -1/a$ is three, since the denominator there is $1 - a^{-2} \neq 0$. Hence ϕ , viewed as a quadratic in b , is irreducible over $\mathbb{Q}(a, y)$. Its two coefficients are coprime in $\mathbb{Q}(a)[y]$, so Gauss's lemma gives irreducibility in $\mathbb{Q}(a)[y, b]$. Viewed instead as a polynomial in y , it is primitive over $\mathbb{Q}(a)[b]$ (its y -coefficient is the unit $3a$ there), and is therefore irreducible over $\mathbb{Q}(a, b)$. Thus $\mathbb{L} = \mathbb{Q}(a, b)[y]/(\phi)$ is a field.

A principal subresultant coefficient is polynomial in the coefficients of its two arguments. If it vanished in $\mathbb{L}$, its polynomial numerator would be divisible by ϕ after clearing coefficient denominators. Primitivity in $\mathbb{Z}[a, b][y]$ follows because the coefficients $3a$ and $1 - b^2$ are coprime over $\mathbb{Q}[a, b]$; Gauss's lemma then makes the divisibility valid before specialization. The coefficient would consequently vanish at every modular root of the specialized ϕ for which denominators survive. The checked specialization preserves the x -degrees and has gcd degree two, so its degree-two principal subresultant coefficient is nonzero. This proves the generic bound $\deg \gcd_{\mathbb{L}}(\hat{E}_{10}, \hat{E}_{13}) \leq 2$. Together with the verified double root $x = \psi$, this identifies the gcd and yields $X_q = 0$, a contradiction.

Branches for group 334.

On the diagonal, $x = y$ first gives $B = \pm A$. The negative sign forces $T = 0$. For the positive sign, put $(A, B, x, y, T) = (z, z, t, t, v)$ and remove the common monomial from each specialized g_i , obtaining P_i . Choose signs of P_1, P_4 so that their v^2 coefficients are $-1, +1$. Define

$$\begin{aligned} Q_1 &= \text{prim Res}_v(P_6, P_4), \\ Q_2 &= \text{prim Res}_z(P_1 + P_4, P_3), \\ Q_3 &= \text{prim Res}_z(Q_1, P_3). \end{aligned}$$

The exact computation gives $0 \neq \text{Res}_t(Q_2, Q_3) \in \mathbb{Z}[a]$, excluding a common zero at generic a . Off the diagonal, let $s = x + y$, $\pi = xy$, and use the swap $(A \ B)(x \ y)$. For $1 \leq i \leq 14$, $i \neq 3, 12$, take the symmetric part $E_i + \sigma E_i$ and the divided antisymmetric part $(E_i - \sigma E_i)/(x - y)$; for $i = 3, 12$, take the norm $E_i \sigma E_i$. Rewrite each expression exactly in (s, π) and verify by back-substitution. The branch factors are

$$\begin{aligned} a\pi^2 - s^2 + 2\pi, \quad a\pi^2 + s^2 - 2\pi, \\ a\pi^3 + s^3 - 3s\pi, \quad a^2\pi^4 - s^4 + 4s^2\pi. \end{aligned}$$

The checker treats their zero loci separately. On the leaf it uses the symmetric and antisymmetric parts of E_{10} and the norm of E_{12} . Write $\hat{E}_{10s}, \hat{E}_{10a}, \hat{E}_{12n}$ for the leaf quotients of the symmetric, divided antisymmetric, and norm polynomials after removing only the nonzero branch factors. Put

$$R_1 = \text{Res}_s(\hat{E}_{10s}, \hat{E}_{10a}), \quad R_2 = \text{Res}_s(\hat{E}_{10s}, \hat{E}_{12n}), \quad P_* = (a^2 - 1)\pi - 3.$$

Remove the a -content and all factors π, P_* from R_1 , leaving R'_1 . Preserved degrees and the coprime specialization $a = 7$ modulo 1000003 give $\text{Res}_\pi(R'_1, R_2) \neq 0$ over $\mathbb{Q}(a)$. Hence a common zero has $\pi = 0$ or $P_* = 0$, and the torus excludes $\pi = 0$. The exact factorization further gives

$$\begin{aligned} \text{prim}_s \gcd(\text{Res}_\pi(\hat{E}_{10s}, P_*), \text{Res}_\pi(\hat{E}_{10a}, P_*)) &= L_*^e, \\ L_* &= (a^2 - 1)s + 3a, \end{aligned}$$

up to a unit in $\mathbb{Q}(a)$, with $e > 0$; prim_s removes parameter content. For $T_* = n_A n_B n_T$ in (s, π) , the checker verifies $L_* \mid \text{Res}_\pi(T_*, P_*)$. Because P_* is linear, $P_* = L_* = 0$ forces $T_* = 0$, which is impossible on the torus. The mass exception $a^2 = 1$ is included when spreading.

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